Manmohan Ganesan

Data Scientist | Machine Learning | RAG | Python

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Summary

Data Scientist with hands-on experience in machine learning, NLP, and Retrieval-Augmented Generation (RAG) systems. Built end-to-end AI projects involving data pipelines, vector databases, and scalable ML workflows using Python, LangChain, SQL, and modern AI tooling.

Skills

Languages
Python, SQL, JavaScript

AI / LLM
LLM Fine-Tuning (QLoRA, peft, trl), LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Vector Databases, FAISS, ChromaDB, Hugging Face, Gemini API, OpenAI API.

Data Analysis & Visualization
Pandas, Numpy, Matplotlib, Seaborn, Tableau, Power BI, Data Cleaning, Data Visualization

Machine Learning
PyTorch, Scikit-learn, Supervised Learning, Regression, Classification, Model Evaluation, Feature Engineering, Data Preprocessing, Exploratory Data Analysis (EDA)

Tools & Technologies
Git, Github, Streamlit, YAML, Logging Frameworks, Config-Driven Pipelines, Jupyter Notebook

Experience

Software Developer, *SysCloud* 2020 – 2021
Chennai

- Analyzed operational datasets using SQL and Excel to support reporting, workflow monitoring, debugging, and issue analysis.
- Developed SQL queries involving JOINs, aggregations, filtering, and subqueries to retrieve, validate, and analyze business-critical data.
- Automated repetitive data-processing and reporting workflows using Python, reducing manual effort and improving operational efficiency.
- Built Excel-based reports and dashboards using pivot tables, lookup functions, and data analysis techniques for operational tracking and reporting.
- Collaborated with cross-functional teams to analyze system logs, investigate recurring issues, and improve operational workflows.

Projects

E-Commerce Intent Prediction LLM, *Personal Portfolio Project* 10/2026 – 11/2026

• Fine-tuned the 7-billion parameter Qwen2.5-Instruct model using 4-bit QLoRA to predict e-commerce user revenue intent, achieving a 78.0% prediction accuracy and a 100.0% valid JSON schema output rate.

• Engineered a memory-constrained training pipeline in PyTorch on Kaggle T4 GPUs using Hugging Face transformers, peft, trl, and bitsandbytes, manually casting adapter parameters and overriding native mixed-precision configurations to resolve VRAM bottlenecks.

• Executed native CPU weight merging of the LoRA adapters and compiled the unified model into a highly optimized 4-bit GGUF format using llama.cpp for localized, resource-efficient inference.

Cart Abandonment Intelligence System, *Personal Portfolio Project* 03/2026 – 03/2026

- Designed and developed an AI-powered cart abandonment intelligence system that predicts user abandonment behavior and generates personalized intervention strategies.
- Built an XGBoost-based behavioral prediction pipeline using engineered session-level features such as engagement score, bounce rate, exit rate, scroll depth, and price sensitivity, achieving ~90% classification accuracy.
- Implemented SHAP explainability and a RAG-based diagnosis pipeline using HuggingFace embeddings and ChromaDB to combine user-level prediction reasoning with population-level behavioral insights.
- Engineered a multi-stage workflow integrating ML prediction, what-if simulation, intervention impact estimation, and LLM-powered recommendation generation using Gemini API.
- Developed a multi-page Streamlit dashboard for prediction, explainability, diagnosis, and strategy generation, demonstrating production-oriented AI system design and decision intelligence workflows.

Google Play Store Analytics Dashboard, *(Power BI)* 01/2026 – 01/2026

- Built an interactive Power BI dashboard analyzing app ratings, installs, reviews, pricing, and category-wise performance trends.
- Performed data cleaning and transformation to derive actionable insights from Google Play Store datasets.
- Designed KPI-driven visualizations to analyze app categories, user engagement, and install trends using Power BI.

Education

Master's in Computer Application, *Christ University* 2018 – 2021
CGPA: 8.73

B.Sc (Physics, Mathematics, Computer Science), *St. Joseph's University* 2015 – 2018
CGPA: 7.5

Certifications

- IIT Patna: Professional Fellowship in Data Science and Agentic AI
- Oracle Certified AI Foundations Associate – 2025
- JPMorgan Chase & Co. - Excel Skills
- Oracle Certified Generative AI Professional – 2025
- Microsoft Certified: Azure Fundamentals 2026
- Full Stack Data Science Program - AlmaBetter
- Google Advanced Data Analytics Professional

Awards

National level Hackathon Winner, *Rajagiri College, Ernakulam* 01/08/2019

Volunteering

Teacher 11/2025 – Present

Bengaluru

- Taught students from 1st to 10th standard across multiple subjects during weekend learning sessions.
- Mentored students individually and in groups to improve academic understanding and personal confidence.
- Collaborated with fellow volunteers to manage classrooms and create an engaging learning environment.